

# DBMS-Module 4

## Question Bank

1. How functional dependencies contribute to the process of database normalization. Illustrate your answer with suitable examples. Analyze the need for normalization in DBMS. Outline 1NF and 2NF with examples.
2. Explain how functional dependencies help in database normalization. How does normalization help in eliminating redundancy?
3. What are functional dependencies? Give examples.
4. Explain First Normal Form (1NF) with an example.
5. Explain Second Normal Form (2NF) with an example.
6. Convert a 3NF relation into BCNF.
7. Explain the steps to compute a minimal cover.
8. Consider 2 sets of functional dependencies. Examine whether they are equivalent.
  - a)  $F = \{A \rightarrow C, AC \rightarrow D, E \rightarrow AD, E \rightarrow H\}$  AND  $G = \{A \rightarrow CD, E \rightarrow AH\}$ .
  - b)  $X = \{AB \rightarrow CD, B \rightarrow C, C \rightarrow D\}$ , AND  $Y = \{AB \rightarrow C, AB \rightarrow D, C \rightarrow D\}$
9. Explain Third Normal Form (3NF) and Boyce–Codd Normal Form (BCNF), and illustrate each with suitable examples.
10. Analyze the following relation schema and its functional dependencies to determine its level of normalization:  
Given a relation  $R(A, B, C, D)$  with functional dependencies  
 $F = \{A \rightarrow B, B \rightarrow C, C \rightarrow D\}$ 
  1. Analyze whether the relation  $R$  is in Third Normal Form (3NF). Justify your answer by examining the functional dependencies and identifying any violations.
  2. If the relation is already in 3NF, analyze and transform it further into Boyce-Codd Normal Form (BCNF) by identifying violating dependencies and performing appropriate decomposition.
11. Examine given set of functional dependencies and construct minimal cover
  - a)  $E: \{AB \rightarrow D, B \rightarrow C, AE \rightarrow B, A \rightarrow D, D \rightarrow EF\}$
  - b)  $F: \{A \rightarrow BC, B \rightarrow C, A \rightarrow B, AB \rightarrow C\}$